# Ancient pollen tells the story of ice age footprints

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Some of the oldest human footprints in North America are at White Sands National Park in New Mexico. A new study adds more proof that they were made during the last **ice age**, about 23,500 to 21,700 years ago.

For a long time, many scientists thought people first came to North America about 16,000 to 12,000 years ago. So these footprints surprised them, and scientists have argued about their age for years.

Researchers from the University of Arizona studied hundreds of thousands of tiny **pollen** grains from the tracks. **Pollen** is a powder made by plants. It showed what plants grew there and what the weather was like.

The land changed fast. In just a few **decades**, evergreen forests turned into **plains** with oak and sage. A giant lake rose and shrank for about 1,500 years and then dried up.

People walked there alongside **mammoths**, giant **sloths** and dire wolves, said scientist Jessica Tierney.

## Word Treasure

**ice age** -- An ice age is a long time when Earth was much colder and big sheets of ice covered land.

**pollen** -- Pollen is a fine powder made by plants.

**decades** -- Decades are periods of 10 years.

**plains** -- Plains are large, flat areas of land with few trees.

**mammoths** -- Mammoths were huge, hairy animals related to elephants that lived long ago.

**sloths** -- Sloths are slow-moving animals, and giant ground sloths lived long ago.

## Background you need

Long ago, during the last ice age, Earth was much colder and many animals that are now gone, like mammoths, were alive.

Footprints pressed into soft mud can harden and stay hidden for thousands of years.

Tiny pollen grains last a long time, so scientists can use them to learn which plants grew in a place long ago.

## Break it down

**WHO:** Researchers from the University of Arizona and scientist Jessica Tierney
**WHAT:** A study of pollen from some of North America's oldest human footprints
**WHERE:** White Sands National Park in New Mexico
**WHY:** To learn how old the footprints are and what the land was like when people walked there

## Why it matters

Clues as tiny as pollen can tell us when people first lived in North America and how they handled big changes in nature.

## Test what you remember

**1. Where are the footprints?**

- (A) Yellowstone
- (B) Alaska
- (C) The Grand Canyon
- (D) White Sands National Park in New Mexico

**2. About how old are the footprints?**

- (A) About 23,500 to 21,700 years
- (B) About 2,000 years
- (C) About 200 years
- (D) About 1 million years

**3. Why were scientists surprised?**

- (A) The footprints were made by dinosaurs
- (B) The park was underwater
- (C) Many thought people came to North America later
- (D) The footprints were in a museum

**4. What is pollen?**

- (A) A kind of rock
- (B) A type of footprint
- (C) A powder made by plants
- (D) Ice from a lake

**5. What did forests turn into in just a few decades?**

- (A) Ice sheets
- (B) Deserts with no plants
- (C) Big lakes
- (D) Plains with oak and sage

**6. Which animals lived there with the people?**

- (A) Polar bears
- (B) Mammoths, giant sloths and dire wolves
- (C) Lions and tigers
- (D) Dinosaurs

Answer key (try the quiz first!): 1: D 2: A 3: C 4: C 5: D 6: B

# Questions to think about

## MY THOUGHTS (at least 20 words):